

HiveForce Labs

THREAT ADVISORY



ATTACK REPORT

Red Eyes Exploits Hangul EPS Vulnerability and Steganography to Spread Malware

Date of Publication

February 16, 2023

Admiralty Code

A1

TA Number

TA2023083

Summary

Attack Begin: January 2023

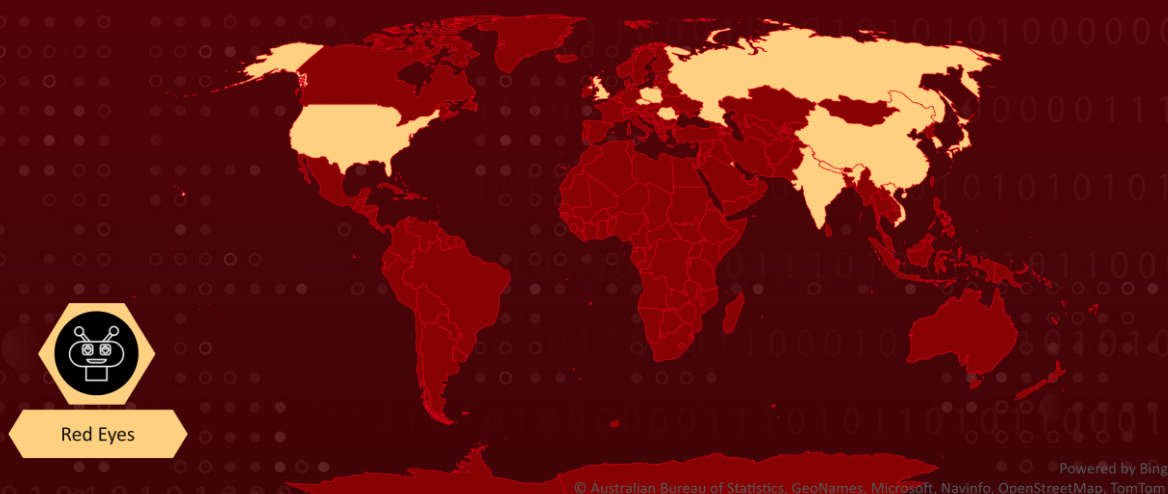
Threat Actor: Red Eyes (Reaper, APT 37, Ricochet Chollima, ScarCraft, Thallium, Group 123, Red Eyes, Geumseong121, Venus 121, Hermit, InkySquid, ATK 4, ITG10)

Attack Countries: China, Czech, Hong Kong, India, Japan, Kuwait, Nepal, Poland, Romania, Russia, South Korea, UK, USA, Vietnam

Attack Sector: Aerospace, Automotive, Chemical, Financial, Government, Healthcare, High-Tech, Manufacturing, Technology, Transportation

Attack: The Red Eyes group used an old vulnerability in Hangul word processor to spread malicious code via steganography, stealing personal PC information and mobile phone data, and executing C&C commands using a shared memory section.

Attack Regions



CVE

CVE	NAME	PATCH
CVE-2017-8291*	Artifex Ghostscript Type Confusion Vulnerability	

* zero-day vulnerability

Attack Details

#1

The Red Eyes is responsible for stealing personal information from both PCs and mobile phones, and in this specific attack, they utilized the Hangul EPS vulnerability and a steganography technique to spread malicious code. The vulnerability used in the attack is an old one that has already been patched, but the attackers were able to exploit it because the target was using an outdated version of the software.

#2

The use of steganography to download malicious code and the registry RUN key registration command for C&C server communication are consistent with previous Red Eyes group attacks. While the group typically uses PowerShell and Chinotto malware for remote control, a new malware was identified in this attack that utilizes a shared memory section to execute C&C commands.

Recommendations



Security Leaders

Phishing simulations and routine education and awareness training and communications rarely account for MFA fatigue and web browser hygiene. Integrate and communicate all lessons learned.



Security Engineers

- **Uni5 Users:** This is an actionable threat advisory in HivePro Uni5. Prioritize and block all indicators attributed to the threat actors and attacks through your Command Center. Test your controls with Uni5's Breach & Attack Simulation.
- **All Engineers:** Refer to and action upon the 'Potential MITRE ATT&CK TTPs' & 'Indicators of Compromise (IoC)' on the following pages.

Potential MITRE ATT&CK TTPs

<u>TA0001</u> Initial Access	<u>TA0002</u> Execution	<u>TA0003</u> Persistence	<u>TA0004</u> Privilege Escalation
<u>TA0005</u> Defense Evasion	<u>TA0006</u> Credential Access	<u>TA0007</u> Discovery	<u>TA0009</u> Collection
<u>TA0010</u> Exfiltration	<u>TA0011</u> Command and Control	<u>T1070</u> Indicator Removal	<u>T1070.004</u> File Deletion
<u>T1574</u> Hijack Execution Flow	<u>T1574.002</u> DLL Side-Loading	<u>T1547</u> Boot or Logon Autostart Execution	<u>T1547.001</u> Registry Run Keys / Startup Folder
<u>T1059</u> Command and Scripting Interpreter	<u>T1059.001</u> PowerShell	<u>T1036</u> Masquerading	<u>T1132</u> Data Encoding
<u>T1041</u> Exfiltration Over C2 Channel	<u>T1055</u> Process Injection	<u>T1113</u> Screen Capture	<u>T1056</u> Input Capture
<u>T1560</u> Archive Collected Data	<u>T1048</u> Exfiltration Over Alternative Protocol	<u>T1027</u> Obfuscated Files or Information	<u>T1140</u> Deobfuscate/Decode Files or Information
<u>T1562</u> Impair Defenses	<u>T1562.003</u> Impair Command History Logging	<u>T1562.001</u> Disable or Modify Tools	<u>T1057</u> Process Discovery
<u>T1016</u> System Network Configuration Discovery	<u>T1071</u> Application Layer Protocol	<u>T1071.001</u> Web Protocols	<u>T1219</u> Remote Access Software
<u>T1566</u> Phishing	<u>T1566.002</u> Spearphishing Link	<u>T1204</u> User Execution	<u>T1203</u> Exploitation for Client Execution
<u>T1105</u> Ingress Tool Transfer	<u>T1106</u> Native API		

✂ Indicator of Compromise (IOCs)

TYPE	VALUE
MD5	8b666fc04af6de45c804d973583c76e0 93c66ee424daf4c5590e21182592672e 7bab405fbc6af65680443ae95c30595d 9083c1ff01ad8fabbcd8af1b63b77e66 4488c709970833b5043c0b0ea2ec9fa9 7f5a72be826ea2fe5f11a16da0178e54

✂ Patch Link

<http://openwall.com/lists/oss-security/2017/04/28/2>

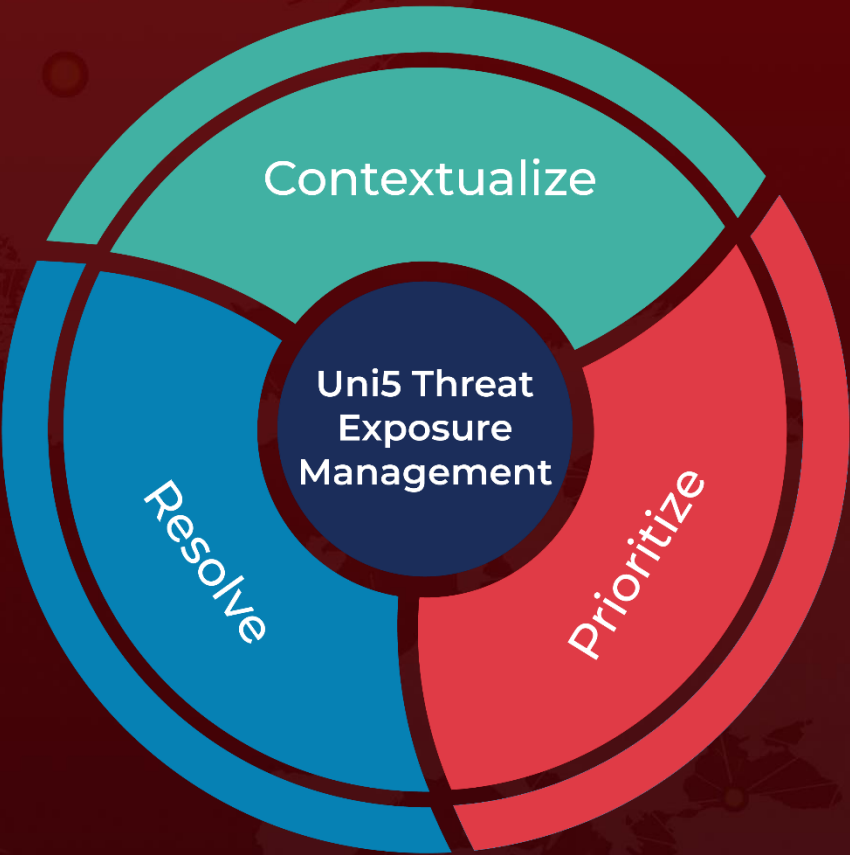
✂ References

<https://asec.ahnlab.com/ko/47622/>

What Next?

At Hive Pro, it is our mission to detect the most likely threats to your organization and to help you prevent them from happening.

Book a free demo with HivePro Uni5: Threat Exposure Management Platform.



REPORT GENERATED ON

February 16, 2023 • 1:00 AM

© 2023 All Rights are Reserved by HivePro



More at www.hivepro.com